

LAPORAN EXCEL ANALYSIS

E-Commerce Sales Dataset — Day 2

Roadmap 7 Hari Data Analyst Portfolio — Dokumentasi Lengkap Langkah demi Langkah

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1. Pendahuluan

Laporan ini mendokumentasikan proses analisis data pada dataset e-commerce sales yang sudah dibersihkan di Day 1 (cleaned_sales_data.xlsx, 3.933 baris). Analisis dilakukan sepenuhnya di Microsoft Excel menggunakan Pivot Table, formula (SUMIFS, COUNTIFS, IF, INDEX/MATCH), dan PivotChart, sesuai checklist Day 2 dalam roadmap Data Analyst portfolio 7 hari.

Berbeda dari Day 1, laporan ini disusun sebagai tutorial langkah demi langkah — setiap bagian menjelaskan CARA melakukan sesuatu di Excel (menu apa yang diklik, field apa yang di-drag, formula apa yang ditulis), bukan cuma menampilkan hasil akhirnya. Tujuannya supaya proses ini bisa diulang lagi atau diajarkan ke orang lain tanpa harus mengingat dari nol.

File kerja: Sales_Analysis.xlsx — dibuat dari Save As atas cleaned_sales_data.xlsx, dengan tambahan sheet pivot_tabels, Summary, Charts, dan Insights.

2. Ringkasan Eksekutif

Tabel 1 merangkum KPI utama hasil analisis.

Metrik	Nilai
Total Revenue (Completed)	\$252.227,72
Total Orders (Completed)	3.220
Average Order Value	\$78,33
Total Revenue (All Status)	\$306.842,22
% Revenue at Risk	17,8%

Tabel 1. KPI utama Sales Performance Summary

3. Tahapan Analisis (Tutorial)

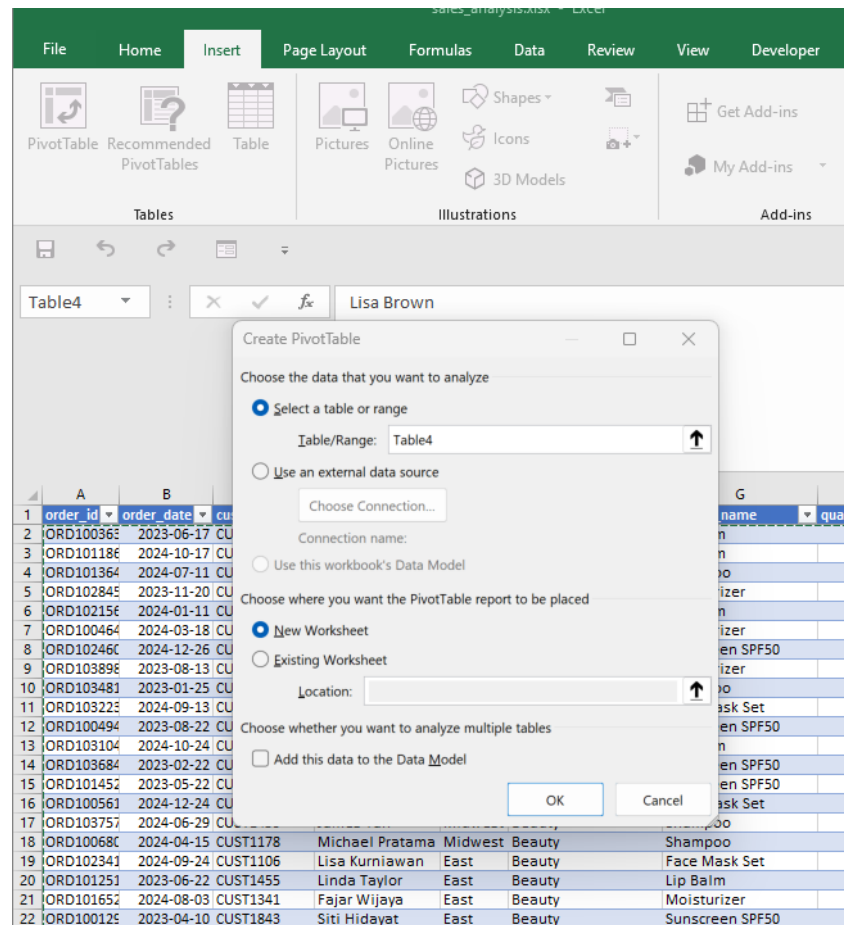
3.1 Membuat Pivot Table Pertama

Semua analisis Top Products, Top Categories, Monthly Revenue, Regional Performance, dan Customer Segments dibangun dari Pivot Table yang merujuk ke Table data di sheet cleaned_sales_data (bernama Table4).

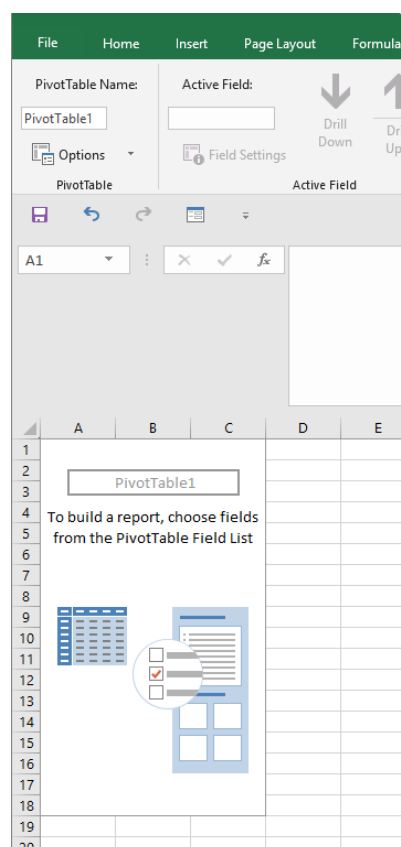
Cara membuatnya:

1. Klik satu cell di dalam Table data (sheet cleaned_sales_data).
2. Insert > PivotTable. Excel otomatis mendeteksi Table/Range (Table4).

3. Pilih New Worksheet (atau Existing Worksheet kalau mau ditaruh di sheet pivot_tabels yang sudah dibuat). Klik OK.



Gambar 1. Dialog Create PivotTable — sumber data Table4, ditempatkan di worksheet baru



Gambar 2. Pivot Table kosong sebelum field ditambahkan — field list muncul di panel kanan

4. Drag field yang diinginkan ke area Rows dan Values di panel PivotTable Fields sebelah kanan. Kombinasi field inilah yang membedakan tiap pivot table (dijelaskan per bagian di bawah).

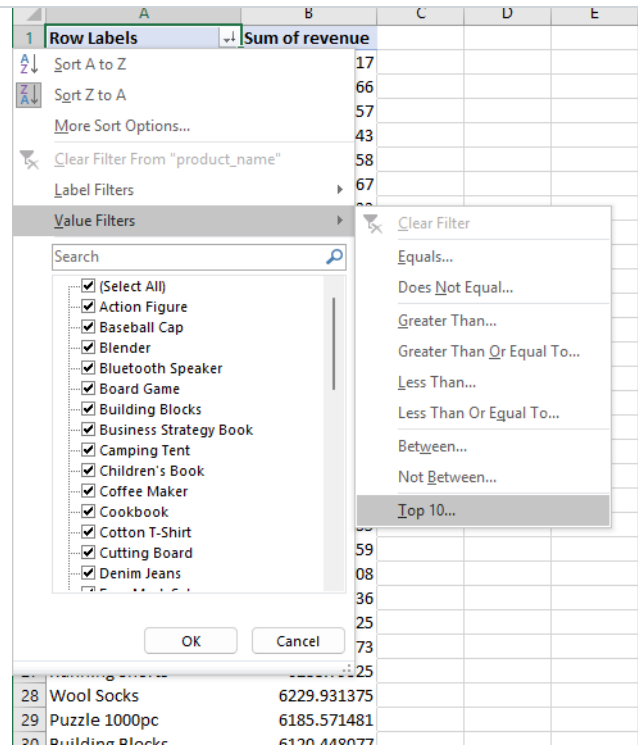
3.2 Top 10 Products by Revenue

1. Drag product_name ke Rows, revenue ke Values (otomatis jadi Sum of revenue).

	A	B
1	Row Labels	Sum of revenue
2	Business Strategy Book	3326.066429
3	Cookbook	3695.323578
4	Self-Help Guide	3742.939464
5	Fiction Novel	3819.53
6	Sunscreen SPF50	4264.666863
7	Shampoo	4515.153604
8	Children's Book	4574.09405
9	Moisturizer	4916.480776
10	Face Mask Set	4987.59
11	Winter Jacket	5152.685789
12	Lip Balm	5217.873
13	Cotton T-Shirt	5616.26
14	Action Figure	5845.642621
15	RC Car	6064.454259
16	Building Blocks	6120.448077
17	Puzzle 1000pc	6185.571481
18	Wool Socks	6229.931375
19	Running Shorts	6288.79325
20	Denim Jeans	6378.182973
21	Hoodie	6480.8525
22	Hiking Backpack	6556.623636
23	Board Game	7024.31708
24	Jump Rope	7076.59
25	Baseball Cap	7138.35
26	Water Bottle	7207.49925
27	Wireless Earbuds	8143.47
28	Coffee Maker	8161.383165
29	Yoga Mat	8185.853125
30	Blender	8188.864648
31	Cutting Board	8415.375753
32	Wireless Mouse	8673.886452
33	Throw Pillow	8870.805172
34	Storage Containers	8998.935556
35	LED Desk Lamp	9069.239643
36	Bluetooth Speaker	9186.68
37	Resistance Bands	9483.796634
38	Laptop Stand	9589.976133
39	Non-Stick Pan	9605.929667
40	USB-C Cable	9894.285758
41	Camping Tent	10146.6343
42	Power Bank	10494.44957
43	Smartphone Case	11621.86466
44	Webcam	11684.86917
45	Grand Total	306842.2194
46		

Gambar 3. Semua produk dengan Sum of revenue, sebelum difilter Top 10

2. Klik kanan salah satu angka revenue → Sort → Sort Largest to Smallest.
3. Klik dropdown filter pada Row Labels → Value Filters → Top 10...



Gambar 4. Menu Value Filters > Top 10 pada dropdown Row Labels

4. Pastikan pengaturan filter “Top 10 Items” berdasarkan Sum of revenue, klik OK.

Top 10 Products by Revenue	
Row Labels	Sum of revenue
Webcam	11684.87
Smartphone Case	11621.86
Power Bank	10494.45
Camping Tent	10146.63
USB-C Cable	9894.29
Non-Stick Pan	9605.93
Laptop Stand	9589.98
Resistance Band	9483.80
Bluetooth Speaker	9186.68
LED Desk Lamp	9069.24
Grand Total	100777.73

Gambar 5. Hasil akhir: Top 10 Products by Revenue, dipimpin Webcam (\$11.684,87)

3.3 Top Categories

Caranya sama seperti Top Products, tapi field Rows diganti category, dan tidak perlu filter Top 10 karena kategori cuma ada 7 total.

1. Buat pivot table baru: category ke Rows, revenue ke Values.
2. Sort Largest to Smallest.

Top Categories	
Row Labels	Sum of revenue
Electronics	79289.48
Home & Kitchen	61310.53
Sports & Outdoors	48657.00
Clothing	43285.06
Toys	31240.43
Beauty	23901.76
Books	19157.95
Grand Total	306842.22

Gambar 6. Top Categories — Electronics memimpin dengan \$79.289,48

3.4 Monthly Revenue Trend & Regional Performance

Monthly Revenue Trend

1. Buat pivot table baru: order_date ke Rows, revenue ke Values.
2. Excel otomatis menawarkan grouping tanggal (dialog Group) begitu order_date di-drag ke Rows. Pilih centang Months dan Years saja (uncheck Quarters kalau ada), supaya hasilnya langsung Year → Month tanpa layer Quarter di tengah.

Regional Performance

1. Buat pivot table baru: region ke Rows. Drag revenue ke Values (Sum), lalu drag order_id ke Values sekali lagi, ubah Value Field Settings-nya jadi Count — supaya dapat 2 metrik sekaligus: total revenue dan jumlah order per region.

Top 10 Products by Revenue		Top Categories	
Row Labels	Sum of revenue	Row Labels	Sum of revenue
Webcam	11684.87	Electronics	79289.48
Smartphone C	11621.86	Home & Kitchen	61310.53
Power Bank	10494.45	Sports & Outdoors	48657.00
Camping Tent	10146.63	Clothing	43285.06
USB-C Cable	9894.29	Toys	31240.43
Non-Stick Pan	9605.93	Beauty	23901.76
Laptop Stand	9589.98	Books	19157.95
Resistance Bar	9483.80	Grand Total	306842.22
Bluetooth Spe	9186.68		
LED Desk Lamp	9069.24		
Grand Total	100777.73		
Monthly Revenue Trend			
Row Labels	Sum of revenue		
2023	157565.78		
Jan	12576.92		
Feb	13438.69		
Mar	16241.61		
Apr	12727.19		
May	14949.47		
Jun	9447.58		
Jul	12913.82		
Aug	14167.22		
Sep	14114.10		
Oct	13900.84		
Nov	12892.05		
Dec	10196.29		
2024	149276.44		
Jan	11348.85		
Feb	11352.15		
Mar	12558.72		
Apr	10886.72		
May	14431.93		
Jun	10747.39		
Jul	12179.76		
Aug	13702.97		
Sep	10987.13		
Oct	13284.38		
Nov	13760.26		
Dec	14036.19		
Grand Total	306842.22		

Gambar 7. Monthly Revenue Trend (Year → Month, tanpa Quarter) berdampingan dengan Top Products & Top Categories

	B	C	D	E	F	G
Top 10 Products by Revenue			Top Categories			
	Row Labels	Sum of revenue		Row Labels	Sum of revenue	
	Webcam	11684.87		Electronics	79289.48	
	Smartphone C	11621.86		Home & Kitche	61310.53	
	Power Bank	10494.45		Sports & Outdc	48657.00	
	Camping Tent	10146.63		Clothing	43285.06	
	USB-C Cable	9894.29		Toys	31240.43	
	Non-Stick Pan	9605.93		Beauty	23901.76	
	Laptop Stand	9589.98		Books	19157.95	
	Resistance Bar	9483.80		Grand Total	306842.22	
	Bluetooth Spe	9186.68				
	LED Desk Lamp	9069.24				
	Grand Total	100777.73				
Monthly Revenue Trend						
	Row Labels	Sum of revenue		Row Labels	Sum of revenue	Count of order_id
	2023	157565.78		East	229821.11	2926
	Jan	12576.92		Midwest	59797.74	784
	Feb	13438.69		South	13856.29	182
	Mar	16241.61		West	3367.08	41
	Apr	12727.19		Grand Total	306842.22	3933
	May	14949.47				
	Jun	9447.58				
	Jul	12913.82				
	Aug	14167.22				
	Sep	14114.10				
	Oct	13900.84				
	Nov	12892.05				
	Dec	10196.29				
	2024	149276.44				
	Jan	11348.85				
	Feb	11352.15				
	Mar	12558.72				
	Apr	10886.72				
	May	14431.93				
	Jun	10747.39				
	Jul	12179.76				
	Aug	13702.97				
	Sep	10987.13				
	Oct	13284.38				
	Nov	13760.26				
	Dec	14036.19				
	Grand Total	306842.22				

Gambar 8. Regional Performance dengan 2 kolom (Sum of revenue dan Count of order_id)

Insight yang langsung kelihatan dari sini: region East mendominasi drastis — \$229.821 dari \$306.842 total revenue (75%), dan 2.926 dari 3.933 order (74%). Region West cuma menyumbang \$3.367 (1%).

3.5 Customer Segments

Berbeda dari 4 pivot table sebelumnya, Customer Segments butuh 2 kolom bantu dulu di sheet data sebelum bisa dibuat pivot table-nya — karena “segmen” bukan kolom yang tersedia langsung di data mentah.

Langkah 1 — Hitung jumlah transaksi per customer (COUNTIFS)

Buat kolom baru customer_order_count di sheet cleaned_sales_data:

```
=COUNTIFS($C$2:$C$3934,[@[customer_id]])
```

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	order_id	order_date	customer_id	customer_name	region	category	product_name	quantity	unit_price	order_status	payment_method	revenue	customer_order_count	
92	ORD102840	2023-06-26	CUST1138	Fajar Anderson	East	Beauty	Sunscreens SPF50	3	9.70	Completed	Debit Card	29.10	7	
128	ORD102068	2024-04-10	CUST1677	Fajar Anderson	Midwest	Sports & Outdoors	Hiking Backpack	2	10.42	Completed	Credit Card	20.84	3	
263	ORD100475	2023-03-04	CUST1097	Fajar Anderson	East	Beauty	Lip Balm	1	12.73	Returned	Gift Card	12.73	6	
716	ORD100513	2023-12-07	CUST1097	Fajar Anderson	East	Books	Self-Help Guide	1	19.51	Completed	Debit Card	19.51	6	
1071	ORD100813	2023-01-25	CUST1677	Fajar Anderson	Midwest	Beauty	Face Mask Set	1	24.50	Completed	Gift Card	24.50	3	
1211	ORD103881	2024-06-25	CUST1097	Fajar Anderson	East	Sports & Outdoors	Hiking Backpack	2	26.14	Cancelled	Debit Card	52.28	6	
1403	ORD101727	2024-07-23	CUST1097	Fajar Anderson	East	Beauty	Face Mask Set	1	28.97	Completed	Gift Card	28.97	6	
1922	ORD100951	2024-02-02	CUST1677	Fajar Anderson	Midwest	Sports & Outdoors	Yoga Mat	2	37.46	Completed	Unknown	74.92	3	
2204	ORD101568	2024-09-11	CUST1138	Fajar Anderson	East	Beauty	Moisturizer	2	42.89	Completed	PayPal	85.78	7	
2356	ORD102309	2024-11-18	CUST1138	Fajar Anderson	East	Sports & Outdoors	Resistance Bands	3	46.25	Completed	Debit Card	138.75	7	
2499	ORD102520	2023-07-10	CUST1138	Fajar Anderson	East	Sports & Outdoors	Camping Tent	1	50.21	Completed	PayPal	50.21	7	
2812	ORD101711	2023-05-19	CUST1138	Fajar Anderson	East	Toys	Building Blocks	4	59.18	Completed	Gift Card	236.72	7	
3619	ORD101572	2023-06-08	CUST1097	Fajar Anderson	East	Home & Kitchen	Non-Stick Pan	1	99.63	Pending	Gift Card	99.63	6	
3655	ORD100066	2024-09-27	CUST1138	Fajar Anderson	East	Home & Kitchen	Coffee Maker	1	102.95	Completed	Debit Card	102.95	7	
3739	ORD101265	2024-10-20	CUST1097	Fajar Anderson	East	Electronics	Wireless Earbuds	1	112.31	Cancelled	Gift Card	112.31	6	

Gambar 9. Kolom customer_order_count — menghitung berapa kali tiap customer_id muncul di seluruh data

Langkah 2 — Klasifikasi New vs Repeat Customer (IF)

Buat kolom baru customer_segment, memakai hasil dari customer_order_count:

=IF([@[customer_order_count]]=1,"New Customer","Repeat Customer")

customer_id	customer_name	region	category	product_name	quantity	unit_price	order_status	payment_method	revenue	customer_order_count	customer_segment
24-06-05 CUST1081	Maria Garcia	East	Toys	Action Figure	1	10.17	Completed	Debit Card	10.17	1	New Customer
24-01-15 CUST1692	Ayu Chen	East	Books	Fiction Novel	1	10.39	Completed	Debit Card	10.39	1	New Customer
24-01-03 CUST1429	Rina Hidayat	East	Books	Children's Book	1	11.54	Canceled	Gift Card	11.54	1	New Customer
24-01-03 CUST1204	Dewi Garcia	West	Beauty	Lip Balm	1	13.03	Returned	Credit Card	13.03	1	New Customer
23-10-10 CUST1792	Linda Sari	Midwest	Books	Business Strategy Boc	1	13.83	Completed	PayPal	13.83	1	New Customer
24-01-15 CUST1208	Sarah Pratama	East	Home & Kitchen	LED Desk Lamp	1	13.97	Completed	PayPal	13.97	1	New Customer
24-06-07 CUST1533	Andi Johnson	East	Sports & Outdoors	Camping Tent	2	16.35	Completed	Credit Card	32.70	1	New Customer
23-12-20 CUST1408	Fajar Wijaya	East	Toys	Board Game	3	16.36	Completed	Debit Card	49.08	1	New Customer
24-01-06 CUST1539	Sarah Salim	East	Books	Cookbook	1	16.91	Completed	Unknown	16.91	1	New Customer
23-08-08 CUST1358	Andi Yusuf	East	Beauty	Lip Balm	2	18.08	Completed	Debit Card	36.16	1	New Customer
24-12-03 CUST1518	Siti Salim	East	Books	Self-Help Guide	1	18.65	Completed	Gift Card	18.65	1	New Customer
24-05-25 CUST1686	Rina Garcia	West	Toys	RC Car	4	21.29	Completed	Credit Card	85.16	1	New Customer
24-08-06 CUST1131	Unknown Customer	West	Beauty	Moisturizer	1	22.77	Canceled	Debit Card	22.77	1	New Customer
24-07-14 CUST1782	Budi Brown	Midwest	Books	Self-Help Guide	1	22.97	Completed	Debit Card	22.97	1	New Customer
23-02-24 CUST1044	David Smith	West	Beauty	Face Mask Set	1	23.36	Canceled	PayPal	23.36	1	New Customer
24-06-22 CUST1371	Sarah Taylor	Midwest	Books	Cookbook	1	23.67	Pending	PayPal	23.67	1	New Customer
23-02-21 CUST1557	Rina Lee	Midwest	Books	Fiction Novel	2	24.08	Completed	PayPal	48.16	1	New Customer
23-06-21 CUST1805	John Chen	South	Home & Kitchen	Storage Containers	2	25.03	Completed	Gift Card	50.06	1	New Customer
24-11-02 CUST1749	Unknown Customer	West	Toys	Board Game	1	25.57	Completed	Gift Card	25.57	1	New Customer
24-09-02 CUST1864	John Anderson	South	Books	Business Strategy Boc	1	27.47	Completed	Credit Card	27.47	1	New Customer
23-03-23 CUST1698	James Johnson	South	Beauty	Face Mask Set	1	27.58	Completed	PayPal	27.58	1	New Customer
23-12-16 CUST1598	Nina Halim	Midwest	Beauty	Shampoo	1	28.02	Completed	Credit Card	28.02	1	New Customer
24-05-22 CUST1706	Putri Hidayat	South	Books	Children's Book	1	28.76	Completed	Credit Card	28.76	1	New Customer
23-09-09 CUST1325	Ayu Kurniawan	East	Clothing	Baseball Cap	1	28.87	Completed	Gift Card	28.87	1	New Customer
24-04-21 CUST1556	Budi Wilson	East	Books	Fiction Novel	1	30.45	Completed	Debit Card	30.45	1	New Customer
24-05-20 CUST1570	Siti Wijaya	East	Beauty	Moisturizer	2	32.73	Completed	Debit Card	65.46	1	New Customer
23-01-13 CUST1767	Andi Smith	West	Toys	Building Blocks	1	34.87	Completed	Debit Card	34.87	1	New Customer
23-12-05 CUST1228	Dewi Santos	Midwest	Sports & Outdoors	Resistance Bands	1	35.69	Completed	PayPal	35.69	1	New Customer
23-05-27 CUST1530	Nina Garcia	East	Beauty	Moisturizer	1	35.83	Completed	Gift Card	35.83	1	New Customer
24-12-19 CUST1489	Nina Salim	Midwest	Clothing	Running Shorts	3	36.29	Completed	Gift Card	108.87	1	New Customer
23-05-04 CUST1326	Sarah Kurniawan	South	Beauty	Lip Balm	1	37.09	Completed	Gift Card	37.09	1	New Customer
24-12-16 CUST1538	Dewi Kurniawan	East	Toys	Action Figure	1	40.42	Completed	Credit Card	40.42	1	New Customer
23-03-19 CUST1727	Nina Brown	Midwest	Sports & Outdoors	Resistance Bands	2	52.35	Completed	Debit Card	104.70	1	New Customer
23-11-12 CUST1083	Ahmad Taylor	East	Clothing	Winter Jacket	1	53.26	Completed	Credit Card	53.26	1	New Customer
23-11-22 CUST1536	Maria Hidayat	West	Clothing	Running Shorts	2	58.04	Completed	Credit Card	116.08	1	New Customer
24-09-24 CUST1797	Ahmad Brown	South	Toys	RC Car	3	58.06	Completed	Debit Card	174.18	1	New Customer

Gambar 10. Formula customer_segment diterapkan — customer dengan 1 transaksi ditandai New Customer, sisanya Repeat Customer

customer_id	customer_name	region	category	product_name	quantity	unit_price	order_status	payment_method	revenue	customer_order_count	customer_segment
24-06-05 CUST1081	Maria Garcia	East	Toys	Action Figure	1	10.17	Completed	Debit Card	10.17	1	New Customer
24-01-15 CUST1692	Ayu Chen	East	Books	Fiction Novel	1	10.39	Completed	Debit Card	10.39	1	New Customer
24-01-03 CUST1429	Rina Hidayat	East	Books	Children's Book	1	11.54	Canceled	Gift Card	11.54	1	New Customer
24-01-03 CUST1204	Dewi Garcia	West	Beauty	Lip Balm	1	13.03	Returned	Credit Card	13.03	1	New Customer
23-10-10 CUST1792	Linda Sari	Midwest	Books	Business Strategy Boc	1	13.83	Completed	PayPal	13.83	1	New Customer
24-01-15 CUST1208	Sarah Pratama	East	Home & Kitchen	LED Desk Lamp	1	13.97	Completed	PayPal	13.97	1	New Customer
24-06-07 CUST1533	Andi Johnson	East	Sports & Outdoors	Camping Tent	2	16.35	Completed	Credit Card	32.70	1	New Customer
23-12-20 CUST1408	Fajar Wijaya	East	Toys	Board Game	3	16.36	Completed	Debit Card	49.08	1	New Customer
24-01-06 CUST1539	Sarah Salim	East	Books	Cookbook	1	16.91	Completed	Unknown	16.91	1	New Customer
23-08-08 CUST1358	Andi Yusuf	East	Beauty	Lip Balm	2	18.08	Completed	Debit Card	36.16	1	New Customer
24-12-03 CUST1518	Siti Salim	East	Books	Self-Help Guide	1	18.65	Completed	Gift Card	18.65	1	New Customer
24-05-25 CUST1686	Rina Garcia	West	Toys	RC Car	4	21.29	Completed	Credit Card	85.16	1	New Customer
24-08-06 CUST1131	Unknown Customer	West	Beauty	Moisturizer	1	22.77	Canceled	Debit Card	22.77	1	New Customer
24-07-14 CUST1782	Budi Brown	Midwest	Books	Self-Help Guide	1	22.97	Completed	Debit Card	22.97	1	New Customer
23-02-24 CUST1044	David Smith	West	Beauty	Face Mask Set	1	23.36	Canceled	PayPal	23.36	1	New Customer
24-06-22 CUST1371	Sarah Taylor	Midwest	Books	Cookbook	1	23.67	Pending	PayPal	23.67	1	New Customer
23-02-21 CUST1557	Rina Lee	Midwest	Books	Fiction Novel	2	24.08	Completed	PayPal	48.16	1	New Customer
23-06-21 CUST1805	John Chen	South	Home & Kitchen	Storage Containers	2	25.03	Completed	Gift Card	50.06	1	New Customer
24-11-02 CUST1749	Unknown Customer	West	Toys	Board Game	1	25.57	Completed	Gift Card	25.57	1	New Customer
24-09-02 CUST1864	John Anderson	South	Books	Business Strategy Boc	1	27.47	Completed	Credit Card	27.47	1	New Customer
23-03-23 CUST1698	James Johnson	South	Beauty	Face Mask Set	1	27.58	Completed	PayPal	27.58	1	New Customer
23-12-16 CUST1598	Nina Halim	Midwest	Beauty	Shampoo	1	28.02	Completed	Credit Card	28.02	1	New Customer
24-05-22 CUST1706	Putri Hidayat	South	Books	Children's Book	1	28.76	Completed	Credit Card	28.76	1	New Customer
23-09-09 CUST1325	Ayu Kurniawan	East	Clothing	Baseball Cap	1	28.87	Completed	Gift Card	28.87	1	New Customer
24-04-21 CUST1556	Budi Wilson	East	Books	Fiction Novel	1	30.45	Completed	Debit Card	30.45	1	New Customer
24-05-20 CUST1570	Siti Wijaya	East	Beauty	Moisturizer	2	32.73	Completed	Debit Card	65.46	1	New Customer
23-01-13 CUST1767	Andi Smith	West	Toys	Building Blocks	1	34.87	Completed	Debit Card	34.87	1	New Customer
23-12-05 CUST1228	Dewi Santos	Midwest	Sports & Outdoors	Resistance Bands	1	35.69	Completed	PayPal	35.69	1	New Customer
23-05-27 CUST1530	Nina Garcia	East	Beauty	Moisturizer	1	35.83	Completed	Gift Card	35.83	1	New Customer
24-12-19 CUST1489	Nina Salim	Midwest	Clothing	Running Shorts	3	36.29	Completed	Gift Card	108.87	1	New Customer
23-05-04 CUST1326	Sarah Kurniawan	South	Beauty	Lip Balm	1	37.09	Completed	Gift Card	37.09	1	New Customer
24-12-16 CUST1538	Dewi Kurniawan	East	Toys	Action Figure	1	40.42	Completed	Credit Card	40.42	1	New Customer
23-03-19 CUST1727	Nina Brown	Midwest	Sports & Outdoors	Resistance Bands	2	52.35	Completed	Debit Card	104.70	1	New Customer
23-11-12 CUST1083	Ahmad Taylor	East	Clothing	Winter Jacket	1	53.26	Completed	Credit Card	53.26	1	New Customer
23-11-22 CUST1536	Maria Hidayat	West	Clothing	Running Shorts	2	58.04	Completed	Credit Card	116.08	1	New Customer
24-09-24 CUST1797	Ahmad Brown	South	Toys	RC Car	3	58.06	Completed	Debit Card	174.18	1	New Customer

Gambar 12. Kolom customer_segment lengkap, difilter untuk menampilkan New Customer saja sebagai verifikasi

Langkah 3 — Buat Pivot Table Customer Segments

1. Buat pivot table baru: customer_segment ke Rows, revenue ke Values (Sum), dan order_id ke Values lagi (Count).

Customer Segments		
Row Labels	Sum of revenue	Count of Orders
New Customer	3427.88	48
Repeat Customer	303414.34	3885
Grand Total	306842.22	3933

Gambar 11. Hasil Customer Segments — Repeat Customer mendominasi 3.885 dari 3.933 order (98,8%)

3.6 Overview: Semua Pivot Table

Kelima pivot table (Top Products, Top Categories, Monthly Revenue Trend, Regional Performance, Customer Segments) disusun berdampingan di sheet pivot_tabels supaya mudah dibaca sebagai satu kesatuan dashboard mini.

Top 10 Products by Revenue		Top Categories	
Row Labels	Sum of revenue	Row Labels	Sum of revenue
Webcam	11684.87	Electronics	79289.48
Smartphone Case	11621.86	Home & Kitchen	61310.53
Power Bank	10494.45	Sports & Outdoors	48657.00
Camping Tent	10146.63	Clothing	43285.06
USB-C Cable	9894.29	Toys	31240.43
Non-Stick Pan	9605.93	Beauty	23901.76
Laptop Stand	9589.98	Books	19157.95
Resistance Bands	9483.80	Grand Total	306842.22
Bluetooth Speaker	9186.68		
LED Desk Lamp	9069.24		
Grand Total	100777.73		

Monthly Revenue Trend		Regional Performance	
Row Labels	Sum of revenue	Row Labels	Sum of revenue
2023	157565.78	East	229821.11
Jan	12576.92	Midwest	59797.74
Feb	13438.69	South	13856.29
Mar	16241.61	West	3367.08
Apr	12727.19	Grand Total	306842.22
May	14949.47		
Jun	9447.58		
Jul	12913.82		
Aug	14167.22		
Sep	14114.10		
Oct	13900.84		
Nov	12892.05		
Dec	10196.29		
2024	149276.44		
Jan	11348.85		
Feb	11352.15		
Mar	12558.72		
Apr	10886.72		
May	14431.93		
Jun	10747.39		
Jul	12179.76		
Aug	13702.97		
Sep	10987.13		
Oct	13284.38		
Nov	13760.26		
Dec	14036.19		
Grand Total	306842.22		

Customer Segments		
Row Labels	Sum of revenue	Count of Orders
New Customer	3427.88	48
Repeat Customer	303414.34	3885
Grand Total	306842.22	3933

Gambar 13. Susunan akhir kelima pivot table di sheet pivot_tabels

3.7 Sales Performance Summary (SUMIFS, COUNTIF)

KPI utama dihitung di sheet Summary, bukan sekadar dari Grand Total pivot table — supaya bisa difilter khusus order berstatus Completed (revenue yang benar-benar terealisasi, bukan termasuk Cancelled/Returned/Pending).

Total Revenue (Completed):
`=SUMIFS(Table4[revenue],Table4[order_status],"Completed")`

Total Orders (Completed):
`=COUNTIF(Table4[order_status],"Completed")`

Average Order Value:
`=D3/D4`

Total Revenue (All Status):

=SUM(Table4[revenue])

% Revenue at Risk:
=1-(\$D\$3/\$D\$7)

Sales Performance Summary		
Total Revenue (Completed)	\$252,227.72	
Total Orders (Completed)	3220	
Average Order Value	\$78.33	
Total Revenue (All Status)	\$306,842.22	
% Revenue at Risk	17.80%	

Gambar 14. Sales Performance Summary — Total Revenue Completed \$252.227,72, AOV \$78,33, Revenue at Risk 17,8%

17,8% revenue “at risk” itu insight kuat — hampir 1 dari 5 dolar potensial hilang gara-gara order berstatus cancel, return, atau pending.

3.8 Customer Lookup Tool (XLOOKUP vs INDEX/MATCH)

Rencana awal memakai XLOOKUP, tapi versi Excel yang dipakai ternyata belum mendukung fungsi tersebut (baru tersedia di Excel 365 / Excel 2021 ke atas — kalau di suggestion box formula cuma muncul XIRR/XNPV/XOR, tandanya XLOOKUP belum ada). Sebagai gantinya dipakai kombinasi INDEX + MATCH, yang secara fungsional setara — mencari nilai berdasarkan kecocokan persis (exact match).

Kalau Excel-mu mendukung XLOOKUP (Microsoft 365 / Excel 2021+)

Cara pakainya lebih ringkas — satu fungsi langsung, tidak perlu digabung dengan fungsi lain:

1. Ketik =XLOOKUP(lalu klik cell D12 (nilai yang dicari, yaitu customer_id yang diketik user).
2. Koma, lalu klik-drag/select seluruh kolom customer_id di Table (Table4[customer_id]) — ini "lookup_array", tempat XLOOKUP mencari kecocokan.
3. Koma, lalu klik-drag/select kolom yang mau ditampilkan hasilnya, misal customer_name (Table4[customer_name]) — ini "return_array".
4. Tutup kurung, Enter.

=XLOOKUP(D12,Table4[customer_id],Table4[customer_name])

Bandingkan dengan versi region, segment, dan total orders — pola formulanya identik, tinggal ganti return array-nya:

```
Region:
=XL00KUP(D12,Table4[customer_id],Table4[region])

Segment:
=XL00KUP(D12,Table4[customer_id],Table4[customer_segment])

Total Orders:
=XL00KUP(D12,Table4[customer_id],Table4[customer_order_count])
```

XLOOKUP secara default sudah exact match (tidak perlu argumen tambahan seperti angka 0 di MATCH), dan bisa mencari ke arah kiri maupun kanan tabel — lebih fleksibel dibanding VLOOKUP klasik.

Versi yang benar-benar dipakai di file ini: INDEX + MATCH

Karena XLOOKUP tidak tersedia, dipakai INDEX + MATCH yang hasilnya identik. Pola formulanya sama untuk keempat field yang ditampilkan Customer Lookup Tool — tinggal ganti kolom yang di-INDEX:

```
Customer Name:
=INDEX(Table4[customer_name],MATCH($D$12,Table4[customer_id],0))

Region:
=INDEX(Table4[region],MATCH($D$12,Table4[customer_id],0))

Segment:
=INDEX(Table4[customer_segment],MATCH($D$12,Table4[customer_id],0))

Total Orders:
=INDEX(Table4[customer_order_count],MATCH($D$12,Table4[customer_id],0))
```

Cara baca formulanya: MATCH mencari POSISI baris di mana customer_id sama dengan D12, lalu INDEX mengambil nilai dari kolom yang dipilih (customer_name / region / customer_segment / customer_order_count) di posisi baris itu. Angka 0 di akhir MATCH wajib ditulis — artinya “cari yang PERSIS sama” (exact match), bukan pencarian mendekati. Tanpa itu, hasil pencarian bisa salah karena customer_id tidak terurut.

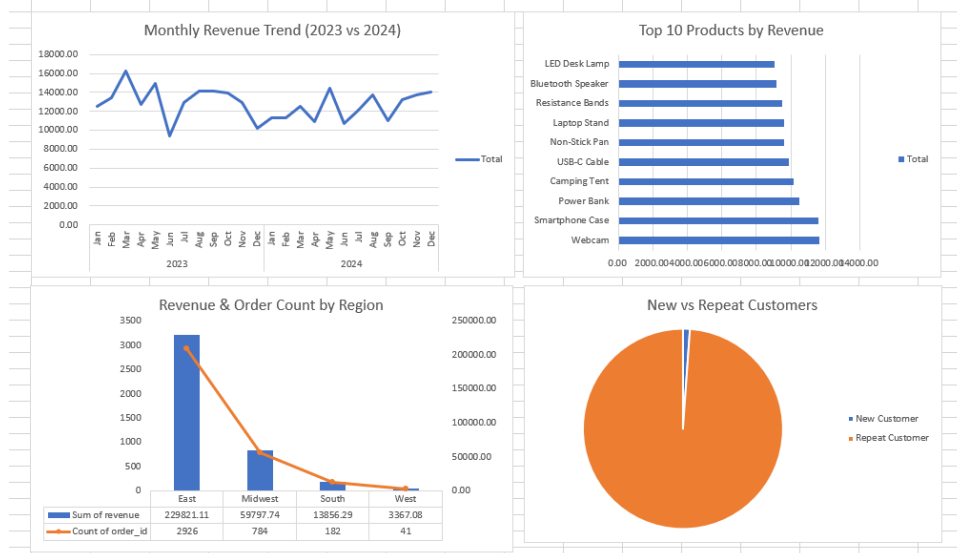
Sales Performance Summary	
Total Revenue (Completed)	\$252,227.72
Total Orders (Completed)	3220
Average Order Value	\$78.33
Total Revenue (All Status)	\$306,842.22
% Revenue at Risk	17.80%
Customer Lookup Tool	
Enter Customer ID	CUST1138
Customer Name	Fajar Anderson

Gambar 15. Customer Lookup Tool — masukkan customer_id (CUST1138), otomatis menampilkan nama customer (Fajar Anderson)

3.9 Charts (PivotChart)

Empat chart dibuat langsung dari pivot table memakai PivotChart (bukan Chart biasa), supaya otomatis update kalau data di pivot table berubah.

1. Klik di dalam pivot table sumber → Insert → PivotChart → pilih tipe chart (Line untuk tren waktu, Bar untuk ranking, Pie untuk proporsi).
2. Untuk chart Regional Performance yang menggabungkan 2 metrik beda skala (revenue dan jumlah order), dipakai Combo Chart: revenue sebagai Column di sumbu utama, Count of order_id sebagai Line di Secondary Axis — supaya dua-duanya kelihatan proporsional, tidak ada yang tertutup skala yang lain.
3. Field Buttons (dropdown filter yang otomatis nempel di pojok chart) disembunyikan lewat PivotChart Analyze > Field Buttons > Hide All, supaya tampilan lebih bersih untuk laporan.



Gambar 17. Dashboard 4 chart: Monthly Revenue Trend, Top 10 Products, Revenue & Order Count by Region, New vs Repeat Customers

3.10 Progress Checklist

12	Day 2	Excel Analysis	Hitung Total Revenue	Excel	SELESAI
13	Day 2	Excel Analysis	Hitung Total Orders	Excel	SELESAI
14	Day 2	Excel Analysis	Hitung Average Order Value	Excel	SELESAI
15	Day 2	Excel Analysis	Cari Top Products	Excel	SELESAI
16	Day 2	Excel Analysis	Cari Top Categories	Excel	SELESAI
17	Day 2	Excel Analysis	Buat Monthly Revenue	Excel	SELESAI
18	Day 2	Excel Analysis	Buat Regional Performance	Excel	SELESAI
19	Day 2	Excel Analysis	Buat Customer Segments	Excel	SELESAI
20	Day 2	Excel Analysis	Pakai Pivot Table	Excel	SELESAI
21	Day 2	Excel Analysis	Pakai XLOOKUP/INDEX MATCH	Excel	SELESAI
22	Day 2	Excel Analysis	Pakai SUMIFS	Excel	SELESAI
23	Day 2	Excel Analysis	Pakai COUNTIFS	Excel	SELESAI
24	Day 2	Excel Analysis	Pakai IF	Excel	SELESAI
25	Day 2	Excel Analysis	Buat charts	Excel	
26	Day 2	Excel Analysis	Tulis 5-10 business insights	Excel	
27	Day 2	Excel Analysis	Simpan sebagai Sales_Analysis.xlsx	Excel	

Gambar 16. Checklist Day 2 pada tahap formula dasar (SUMIFS, COUNTIFS, IF) sudah selesai, Charts dan Insights menyusul

4. Business Insights

Delapan insight berikut disusun berdasarkan seluruh angka dan pola yang ditemukan sepanjang analisis di atas.

1. Ketergantungan Ekstrem pada Region East

Region East menyumbang 75% dari total revenue (\$229.821 dari \$306.842) dan 74% dari total order (2.926 dari 3.933), sementara West cuma menyumbang 1% (\$3.367, 41 order). Ketimpangan seekstrem ini menunjukkan bisnis sangat bergantung pada satu wilayah — risiko besar kalau performa East menurun.

2. 17,8% Revenue Berisiko Tidak Terealisasi

Dari total revenue kotor \$306.842, hanya \$252.228 (82,2%) berasal dari order berstatus Completed. Sebanyak 17,8% (\$54.614) berasal dari order Cancelled, Returned, atau Pending — revenue yang belum tentu benar-benar terealisasi menjadi kas.

3. Basis Pelanggan Didominasi Repeat Customer, Akuisisi Baru Lemah

98,8% order (3.885 dari 3.933) berasal dari Repeat Customer, hanya 48 order dari New Customer. Ini menunjukkan retensi pelanggan sangat kuat, tapi di sisi lain menandakan akuisisi pelanggan baru yang sangat lemah dan perlu ditingkatkan.

4. Electronics Adalah Kategori dan Produk Dominan

Electronics adalah kategori dengan revenue tertinggi (\$79.289, 26% dari total), diikuti Home & Kitchen (\$61.311). Tiga dari lima produk terlaris (Webcam, Smartphone Case, Power Bank) juga berasal dari kategori Electronics — kategori ini jadi pendorong utama penjualan.

5. Books Jadi Kategori dengan Performa Terlemah

Books adalah kategori dengan revenue terendah (\$19.158), kurang dari seperempat revenue Electronics. Kategori ini berpotensi dievaluasi ulang dari sisi variasi produk atau strategi promosi.

6. Tidak Ada Pola Musiman yang Konsisten

Revenue bulanan berfluktuasi tanpa pola musiman yang jelas — puncak di Maret 2023 (\$16.242), lalu turun ke titik terendah di Juni 2023 (\$9.448). Ketidakstabilan ini menunjukkan revenue lebih dipengaruhi faktor acak/operasional dibanding tren musiman.

7. Revenue Tahunan Sedikit Menurun, Bukan Tumbuh

Total revenue turun tipis dari \$157.566 (2023) menjadi \$149.276 (2024), penurunan sekitar 5,3% year-over-year. Meski kecil, ini tetap tren negatif yang perlu diperhatikan, bukan pertumbuhan.

8. AOV Tergolong Rendah, Peluang Bundling/Cross-Sell

Average Order Value hanya \$78,33 dari 3.220 order Completed, konsisten dengan pola quantity yang mayoritas cuma 1 unit per order. Ada peluang menaikkan AOV lewat strategi bundling atau cross-selling produk pelengkap.

5. Ringkasan Checklist Day 2

Task	Status	Keterangan
Hitung Total Revenue	Selesai	SUMIFS, filter order_status = Completed
Hitung Total Orders	Selesai	COUNTIF, filter order_status = Completed
Hitung Average Order Value	Selesai	Total Revenue ÷ Total Orders
Cari Top Products	Selesai	Pivot Table + filter Top 10
Cari Top Categories	Selesai	Pivot Table, 7 kategori
Buat Monthly Revenue	Selesai	Pivot Table, grouping Year + Month
Buat Regional Performance	Selesai	Pivot Table + Secondary Axis chart
Buat Customer Segments	Selesai	Helper column COUNTIFS + IF, lalu Pivot Table
Pakai Pivot Table	Selesai	5 pivot table dipakai total
Pakai XLOOKUP / INDEX-MATCH	Selesai	Diganti INDEX/MATCH — versi Excel tidak mendukung XLOOKUP
Pakai SUMIFS	Selesai	Total Revenue (Completed)
Pakai COUNTIFS	Selesai	customer_order_count
Pakai IF	Selesai	customer_segment
Buat charts	Selesai	4 PivotChart di sheet Charts
Tulis 5-10 business insights	Selesai	8 insight di sheet Insights
Simpan sebagai Sales_Analysis.xlsx	Selesai	3.933 baris data bersih

Tabel 2. Status seluruh task Day 2 Excel Analysis

6. Kesimpulan

Analisis Day 2 menghasilkan Sales_Analysis.xlsx berisi 5 pivot table, 4 chart interaktif, KPI summary dengan formula SUMIFS/COUNTIF, Customer Lookup Tool dengan INDEX/MATCH, dan 8 business insight yang didukung angka konkret. Seluruh proses dibangun di atas dataset bersih 3.933 baris dari Day 1, sehingga angka yang dilaporkan konsisten dengan tahap-tahap sebelumnya.

Temuan paling signifikan: ketergantungan ekstrem pada region East (75% revenue), potensi kehilangan 17,8% revenue akibat order yang tidak Completed, dan akuisisi pelanggan baru yang sangat lemah (cuma 1,2% dari total order). Tiga poin ini jadi kandidat kuat untuk dibahas lebih dalam di Day 6 (Business Case Study).

File Sales_Analysis.xlsx ini menjadi input untuk Day 3 (SQL queries), Day 4 (Python EDA), dan Day 5 (Tableau dashboard) — khususnya angka-angka KPI dan insight di atas akan dipakai sebagai pembandingan/validasi silang di tahap-tahap tersebut.